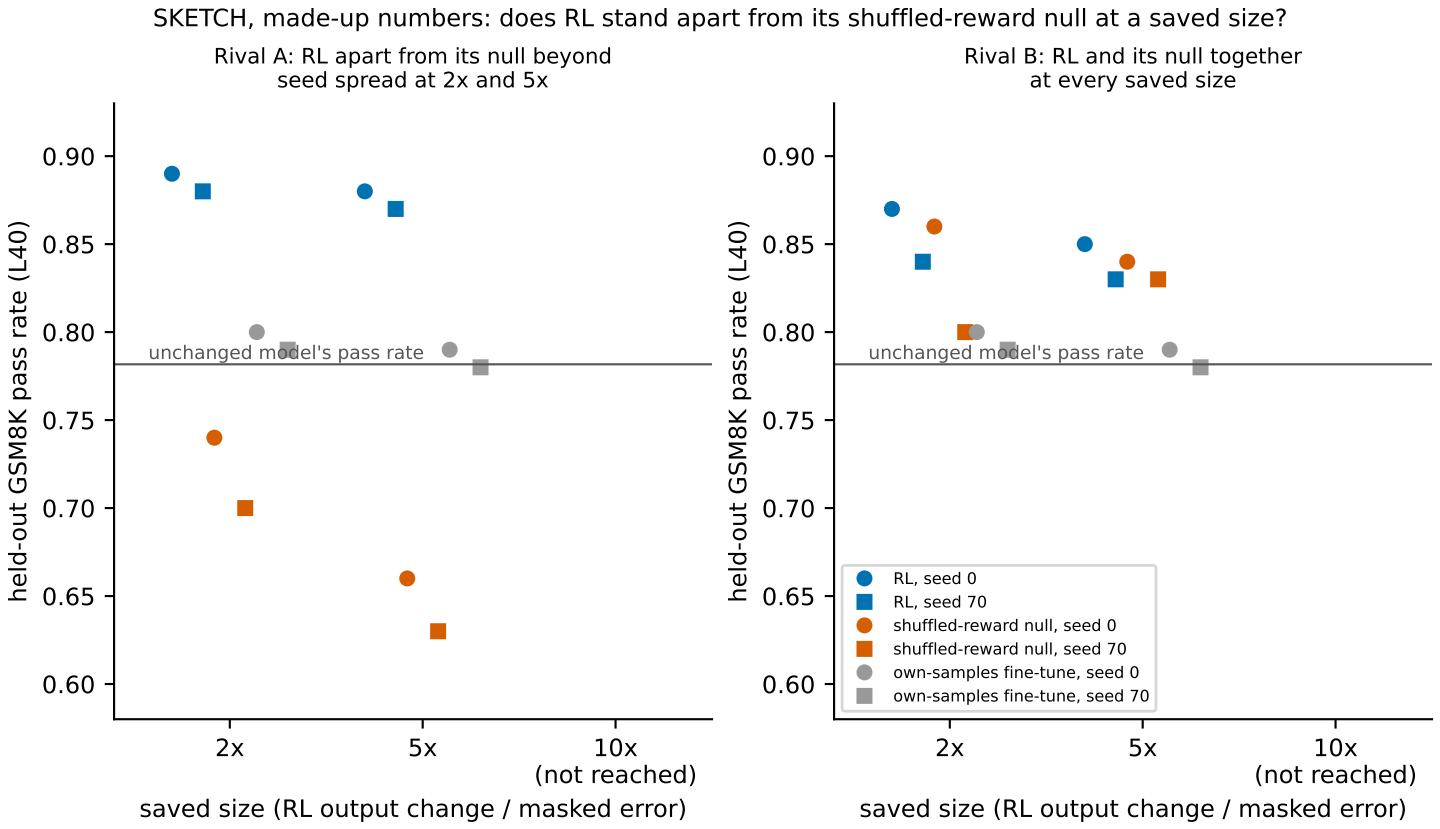
SKETCH, made-up numbers: does RL stand apart from its shuffled-reward null at a saved size?
Rival A: RL apart from its null beyond seed spread at 2x and 5x
Rival B: RL and its null together at every saved size
held-out GSM8K pass rate (L40)
0.90
0.85
0.80
0.75
0.70
0.65
0.60
unchanged model's pass rate
2x
5x
10x (not reached)
saved size (RL output change / masked error)
RL, seed 0
RL, seed 70
shuffled-reward null, seed 0
shuffled-reward null, seed 70
own-samples fine-tune, seed 0
own-samples fine-tune, seed 70
made-up numbers; the run: two seeds per arm, one state per arm, seed and size, pass rate over the ~1250 of 1319 GSM8K test problems that fit